

JEE MAIN REVISION MASTER: DIFFERENTIATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Evaluation of Limits & L'Hopital's Rule	31 PYQs	Limits
Moderate	Continuity & Differentiability at a Point	7 PYQs	Continuity & Differentiability

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, $\lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)}$ (for 1^∞)
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$, $\text{Differentiability} \implies \text{Continuity}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Evaluation of Limits & L'Hopital's Rule (Very High Yield — 31 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The second order derivative of $\cos^{-1}(4x^3-3x)$ with respect to $\cos^{-1}(2x^2-1)$, where $(1/2) < x < 1$ is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If $f(x) = \sin^{-1}(\frac{2x+1}{x^2+1})$, then $f'(1/2) = \dots$ " [Ans: A]
- [2025 Shift] Q#1 (MCQ): "If $y = (\cos x / (1 + \sin x))$, then (a) $(dy/dx) = (-1/(1 + \sin x))$ (b) $(dy/dx) = (1/(1 + \sin x))$ (c) $\frac{dy}{dx} = -\frac{1}{2}$..." [Ans: N/A]

3.2 Continuity & Differentiability at a Point (Moderate Yield — 7 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "If $y = \sqrt[3]{3}$, then $(dy/dx) = \dots$ " [Ans: N/A]
- [2026 Shift] Q#4 (MCQ): "If $\sqrt[3]{(x)\sqrt[3]{3}} = (x+y)^n$ and $x(dy/dx) - y = 0$, then $n = \dots$ " [Ans: C]
- [2021 Shift] Q#1 (MCQ): "For constant a , $(d/dx)(x^a x + x^a a + a^a x + a^a a)$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The second order derivative of $\cos^{-1}(4x^3 - 3x)$ with respect to $\cos^{-1}(2x^2 - 1)$, where $(1/2) < x < 1$ is

- (A) 0
- (B) $-\frac{1}{\sqrt{1-x^2}}$
- (C) $(3/2)$
- (D) $-(3/2)$

Correct Answer: None

Solution Analysis: Take $y = \cos^{-1}(4x^3 - 3x)$ and $t = \cos^{-1}(2x^2 - 1)$ We need (d^2y/dt^2) Since $(1/2) < x < 1$ where $0 < \theta < (\pi/3)$ Now use the standard identities: $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ and $2\cos^2\theta - 1 = \cos 2\theta$ So, $y = \cos^{-1}(\cos 3\theta)$ Since $0 < 3\theta < \pi$, we get $y = 3\theta$ Also, $t = \cos^{-1}(\cos 2\theta)$ Since $0 < 2\theta < \pi$, we get $t = 2\theta$ Thus, $y = 3\theta$, $t = 2\theta$ So, $y = (3/2)t$ Hence, $(dy/dt) = (3/2)$ and $(d^2y/dt^2) = 0$ So the correct o

Benchmark Problem 2: KCET 2026 (Q#3)

If $y = \sqrt[3]{3}$, then $(dy/dx) =$

- (A) $\frac{\tan x}{3y^2 - 1}$
- (B) $\frac{\sec^2 x}{3y - 1}$
- (C) $(\tan x / 3y - 1)$
- (D) $\frac{\sec^2 x}{3y^2 - 1}$

Correct Answer: None

Solution Analysis: Given, $y = \sqrt[3]{3}$ To remove the cube root, cube both sides: $y^3 = \tan x + y$ Now differentiate both sides with respect to x . For the left side, using chain rule: $(d/dx)(y^3) = 3y^2(dy/dx)$ For the right side: $(d/dx)(\tan x + y) = \sec^2 x + (dy/dx)$ So, $3y^2(dy/dx) = \sec^2 x + (dy/dx)$ Bring the terms containing (dy/dx) together: $3y^2(dy/dx) - (dy/dx) = \sec^2 x$ Take (dy/dx) common: $(3y^2 - 1)(dy/dx) = \sec^2 x$ Hence, $(dy/dx) = \frac{\sec^2 x}{3y^2 - 1}$ So the correct option is: Option D